AI-Powered Computer Vision System for Automated Classification and Cataloging of Textile Waste.

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Project Proposal Report

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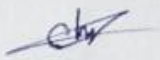
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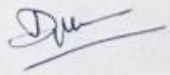
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Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

Large amounts of fabric waste are produced by Sri Lanka's textile and clothing sector throughout the cutting, manufacturing, and quality control phases. This garbage is now sorted and classified by hand, which is a labor-intensive, inconsistent, and unscalable method that prevents the development of a standardized digital inventory necessary for effective upcycling and redistribution. By developing an AI-powered computer vision system for the automated categorization and cataloging of textile waste, our research addresses this crucial bottleneck. Using deep learning, specifically Convolutional Neural Networks (CNNs) with transfer learning, the proposed system analyzes photos of waste fabric and accurately predicts three key characteristics: fabric type (such as cotton or polyester), color, and quality grade. To train the algorithm and achieve a classification accuracy of over 85%, a bespoke dataset of fabric waste photos representative of Sri Lankan industry contexts will be created and enhanced.

A scalable Fast API microservice that can accept picture inputs and produce structured JSON output with predictions and confidence ratings will be used to implement the trained model. By converting unstructured garbage into searchable, standardized digital assets, this module acts as the fundamental "eyes" of the broader digital marketplace ecosystem. The technology facilitates accurate valuation, enables transparent inventory management, and lays the foundation for a data-driven circular economy in Sri Lanka's textile industry by automating the categorization process.

Keywords: Computer Vision, Textile Waste Classification, Convolutional Neural Networks (CNN), Transfer Learning, Deep Learning, Fabric Recognition, Automated Cataloging, Fast API, Circular Economy, Sri Lanka

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Table of Contents

Declaration ..... **Error! Bookmark not defined.**

Abstract 2

Acknowledgement 4

List of Tables 7

List of Figures 8

List of abbreviations 9

1. Introduction 10

   1.1 Background and Context 10

   1.2 Literature Review 12

   1.3 Research Gap 13

2. Objectives 14

   3.1 Main Objective 14

   3.2 Specific Objectives 14

3. Methodology 15

   3.1 Research Approach 15

   3.2 Dataset Creation and Augmentation 15

   3.3 CNN Architecture and Transfer Learning 17

   3.4 Model Training, Validation, and Testing 18

   3.5 Model Evaluation Metrics 18

   3.6 API Development and Deployment 19

4. High-level system architecture 20

5. Requirements 21

   5.1 Stakeholders 21

   5.3 Functional Requirements 21

1. Non-Functional Requirements 23
5.4 System Requirements 24
5.5 User Requirements 24
5.6 Use Cases 25
1. Wireframes 28
6. Commercialization Plan 30
6.1 Target Market 30
6.2 Value Proposition: 30
7. Budget and budget justification 32
8. Work breakdown structure 34
9. Gantt chart 36
Appendices 40

List of Tables

Table 1: Target Classification Categories Table. 15
Table 2: Functional Requirements Table. 21
Table 3: Non-Functional Requirements Table. 23
Table 4: Use Case 1 – Upload Waste Image for Classification Table. 25
Table 5: Use Case 2 – Review and Correct AI Classification Table. 26
Table 6: Task Table 37
Table 7: Work Breakdown Table ..... **Error! Bookmark not defined.**
Table 8: Budget Table. ..... **Error! Bookmark not defined.**

List of Figures

Figure 1: Response in Jason Format. 19
Figure 2: System Overview Diagram – Fabric Classification Module. 20
Figure 3: Wireframe – Image Upload Interface for Sellers. 28
Figure 4: Wireframe – Classification Results Display. 29
Figure 5: Project Gantt Chart. 26

List of abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
EC2	Elastic Compute Cloud
FR	Functional Requirement
GCP	Google Cloud Platform
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
IP	Intellectual Property
JSON	JavaScript Object Notation
JWT	JSON Web Token
NFR	Non-Functional Requirement
OS	Operating System
PIL	Python Imaging Library
ReLU	Rectified Linear Unit
REST	Representational State Transfer
SaaS	Software as a Service
SDG	Sustainable Development Goal
TIM	Technology Integration and Management
TLS	Transport Layer Security
UC	Use Case
VM	Virtual Machine
VPC	Virtual Private Cloud

1. Introduction

1.1 Background and Context

One of the biggest industrial industries in the world, the textile and garment industry, produces a significant quantity of trash throughout production. Global sustainability statistics state that millions of tons of textile waste are produced annually, most of which is dumped in landfills even though it could be recycled, reused, or upcycled. [1, 2]. Due to the serious economic and environmental problems this has caused, businesses and legislators are encouraging circular economy methods that prioritize resource reuse and waste reduction. [3].

One of Sri Lanka's biggest export industries and a major contributor to the country's economy is the clothing industry. [4]. However, significant volumes of post-industrial textile waste, such as fabric offcuts, faulty materials, and useless remnants, are produced by clothing production operations, such as fabric cutting, trimming, and quality checking. Improving sustainability and operational effectiveness in the sector has made effective waste management more crucial. [5].

Most clothing manufacturers now use manual textile waste sorting, in which employees classify waste materials after visually identifying the type, colour, and quality of the fabric. This procedure is inconsistent, subjective, and time-consuming. [6]. The growth of digital circular economy markets, where manufacturers might sell or redistribute textile waste to designers, recyclers, and upcyclers, is further hampered by the lack of trustworthy waste classification. [7].

Potential answers to this problem can be found in recent advances in artificial intelligence (AI) and computer vision. Visual patterns, including colour, texture, and structure, may be automatically identified by computer vision systems. [8]. These systems can learn to identify material properties with high accuracy by using deep learning models like Convolutional Neural Networks (CNNs) [9].

This study suggests creating an AI-based system for classifying textile waste that can automatically determine the kind, colour, and quality of the fabric from pictures of the garbage. To facilitate standardized digital inventories and effective textile waste redistribution, the system will operate as a microservice that interfaces with the suggested TexTread circular marketplace platform.

The study supports Sustainable Development Goal (SDG) 12, Responsible Consumption and Production of the United Nations. [10]. Additionally, the project is part of the Technology Integration and Management (TIM) research cluster since it uses digital platform technologies, computer vision, and artificial intelligence to promote sustainable waste management in the textile sector.

1.2 Literature Review

Automated material recognition systems that increase waste management process efficiency have been made possible by recent developments in computer vision. Recycling plants have made extensive use of vision-based sorting devices to separate items, including paper, metals, and plastics. [11]. When compared to traditional approaches, these systems greatly increase sorting speed and accuracy by automatically classifying materials and identifying visual features using image processing algorithms and machine learning techniques. [12].

Researchers have investigated a number of methods for identifying fabrics in the textile industry. It has been demonstrated that hyperspectral imaging, which records spectral fingerprints across several wavelengths, can differentiate between synthetic fibers like polyester and natural fibers like cotton. [13]. However, the feasibility of hyperspectral systems for many industrial applications is limited by the need for costly sensors, regulated illumination conditions, and specialized equipment. [14].

CNNs are now the most advanced method for image classification jobs due to the quick development of deep learning technology. Strong performance in extracting hierarchical visual characteristics from complicated picture data has been shown by well-known designs like ResNet, VGG, and EfficientNet. [15]. CNN-based methods have been used in several studies to address textile-related issues, including defect identification and fabric texture classification. [16, 17].

Despite these developments, a significant drawback of current research is that the majority of datasets include clean, flat, and well-organized fabric samples that were taken in a controlled laboratory setting. [18]. Real-world textile trash, on the other hand, seems asymmetrical, folded, crumpled, and occasionally partially ruined. The requirement for actual trash-specific datasets is highlighted by the fact that models trained on ideal datasets frequently perform badly on post-industrial textile waste photos. [19].

Another crucial method that reduces training time while increasing classification accuracy is transfer learning, which enables models learned on massive datasets like ImageNet to be modified for new applications [20]. Nevertheless, there hasn't been much study done on using transfer learning methods to classify textile waste in circular economy platforms. [21].

1.3 Research Gap

Even though computer vision methods for textile analysis have been studied in the past, there are still several significant gaps in the management of textile waste, especially in poor nations like Sri Lanka.

1. Instead of classifying textile waste, most current CNN-based models are made for fabric quality inspection or defect identification [22].
2. Localized datasets that reflect the textile waste produced by Sri Lankan clothing factories are few [23].
3. These categorization algorithms are infrequently incorporated into useful circular economy platforms in current research [24].
4. The development of standardized digital inventories required for transparent and scalable trash exchange networks is hindered by the ongoing dependence on manual textile waste sorting [25].

By creating a CNN-based classification model trained on Sri Lankan textile waste data and implementing it as an integrated microservice within the TexTread circular economy platform, this study seeks to overcome these constraints.

2. Objectives

3.1 Main Objective

To create a computer vision system using artificial intelligence (AI) that can recognize and categorize textile waste items automatically to facilitate a digital circular textile marketplace.

3.2 Specific Objectives

1. To evaluate the Sri Lankan garment industry's present textile waste sorting procedures and pinpoint the shortcomings of manual categorization techniques.
2. To create an image-based classification model that can recognize textile waste attributes, including fabric kind, colour, and quality.
3. To use common machine learning assessment measures to assess the suggested AI classification model's performance.
4. To create a scalable system component capable of integrating a digital textile waste marketplace platform with the categorization model.
5. To illustrate how automated sorting of textile waste may enhance waste tracking and promote circular economy principles.

3. Methodology

3.1 Research Approach

In order to create and assess an AI-based system for classifying textile waste, this study combines design science methodology with an experimental machine learning technique. In order to integrate with the TexTread platform, the study entails creating datasets, developing models using deep learning techniques, assessing performance, and deploying the model as a microservice.

3.2 Dataset Creation and Augmentation

- **Data Collection:** Fabric waste samples from nearby tailor shops, small clothing manufacturers, and textile suppliers in the Colombo region will be gathered to build a main dataset. To replicate actual user uploads, these examples will be taken with different lighting, perspectives, and backdrops. To make sure the model is resilient to typical input quality, pictures will be taken with a regular smartphone camera.
- **Data Labelling:** The researcher will manually classify each image with the three characteristics listed in Table 1. To guarantee uniformity, a precise labelling manual will be developed.

Table 1: Target Classification Categories Table.

Attribute	Categories	Notes
Fabric Type	Cotton, Polyester, Nylon, Viscose, Blend	"Blend" for common mixtures like Poly-Cotton.
Color	Red, Blue, Green, Yellow, Black, White, Other	Dominant color. "Other" for multi-colored or complex prints.
Quality Grade	High, Medium, Low	Based on visual assessment of tears, stains, fading, and overall integrity. A rubric will be defined.

- **Target Dataset Size:** Aim for at least 5,000 labelled photos that are distributed fairly evenly throughout the categories. This might be accomplished by gathering between 1,500 and 2,000 distinct photos and using augmentation.
- **Data Augmentation:** Real-time augmentations will be applied during training using the Keras ImageDataGenerator or a comparable package to artificially increase the dataset and enhance model generalization. These augmentations include:
    - Rotation at random (±20 degrees)
    - Changes in height and width
    - Flips, both horizontal and vertical
    - A random Zoom
    - Adjusting brightness

- **Data Splitting:** Three subsets will be created from the final dataset:
    - Training Set (70%): The model is trained using this set.
    - Validation Set (15%): Used to adjust hyperparameters and keep an eye out for overfitting when training.
    - Test Set (15%): Only utilized for the final assessment of performance on unseen data; completely withheld until the model is finalized.

3.3 CNN Architecture and Transfer Learning

- **Basis Architecture:** The basis will be a contemporary, well-known CNN architecture. Among the candidates are:
    - ResNet50: A tried-and-true design that enables training extremely deep networks thanks to residual connections. a solid, trustworthy foundation.
    - EfficientNetB3: A more modern design that achieves cutting-edge accuracy with higher computing efficiency. The best option if processing power permits.
- **Transfer Learning Approach:**

1.4 Load the selected base model, omitting its top classification layer, that has been pre-trained on the ImageNet dataset.

2.4 To preserve the broad visual characteristics that the basic model has learnt, freeze the weights of its first layers.

3.4 Expand the base model with additional trainable layers. A Global Average Pooling layer to reduce the feature maps to a single vector per picture will make up this "head."

- To learn task-specific feature combinations, a few dense (completely linked) layers with ReLU activation are used.
- Three distinct output layers with the corresponding activation function, one for each attribute:
    - Softmax (multi-class classification) for fabric type.
    - Softmax (multi-class categorization) for colour.
    - Softmax or sigmoid (multi-class or ordinal regression, to be determined) for quality grade.
- By using common visual characteristics, this multi-output method enables the single model to predict all three properties at once.

3.4 Model Training, Validation, and Testing

- **Environment:** Python with TensorFlow/Keras or PyTorch will be used for training. To speed up training, if available, a cloud GPU instance (such as Google Colab Pro or AWS EC2 with GPU) will be utilized.
- **Training Procedure:**
    1. The model will be constructed using distinct metrics (accuracy) and loss functions for every output (such as categorical cross-entropy).
    2. The training set with the frozen basic layers will be used to train the model. The ImageNet characteristics will be mapped to our fabric properties via the new "head" layers.
    3. To better fit the dataset, the basic model's top layers may be unfrozen after initial training, and the entire model may be fine-tuned at a very low learning rate.
    4. To avoid overfitting, training will be halted if the validation loss does not improve for several epochs (early stopping). The validation set will be used to track performance after each epoch.
- **Evaluation:** On the untested test set, the final, top-performing model will be assessed once. The following will be the main assessment metrics:
    - **Overall Accuracy:** The proportion of pictures in which each of the three anticipated characteristics is accurate.
    - **Per-Attribute Accuracy:** Precision for each fabric type, colour, and quality level separately.
    - **F1-Score, Precision, and Recall:** For every class inside every attribute to comprehend performance on certain categories (e.g., does the model struggle with "Blend" fabrics?).

3.5 Model Evaluation Metrics

>85% total accuracy on the exam set is the main objective. A detailed picture of the model's advantages and disadvantages will be provided by the per-attribute and per-class metrics.

3.6 API Development and Deployment

- **Framework:** Because of its excellent speed, automated OpenAI (Swagger) documentation, and user-friendliness, Fast API will be utilized to build the microservice.
- **Endpoint:** /classify_fabric will be the only POST endpoint built.
- **Request:** An image file upload (multipart/form-data) will be accepted by the endpoint. JPG and PNG are supported formats.
- **Processing:**
  1. After receiving the image, it will be verified.
  2. It will be normalized and pre-processed to the CNN's needed input size, such as 224x224 pixels.
  3. The loaded TensorFlow/Keras model will get the pre-processed picture for inference.
  4. The model's predictions will be post-processed to provide confidence ratings and labels that are readable by humans.
- **Response:** A JSON object with the following contents will be returned by the API:

```
json
{
  "fabric_type": "Cotton",
  "fabric_type_confidence": 0.95,
  "color": "Blue",
  "color_confidence": 0.98,
  "quality_grade": "High",
  "quality_grade_confidence": 0.82,
  "processing_time_ms": 450
}
```

Figure 1: Response in Jason Format.

- **Deployment**: to guarantee consistency across environments, the Fast API application will be containerized using Docker, deployed on a cloud virtual machine (e.g., AWS EC2, Google Cloud Run), and made available over a secure API gateway.

4. High-level system architecture

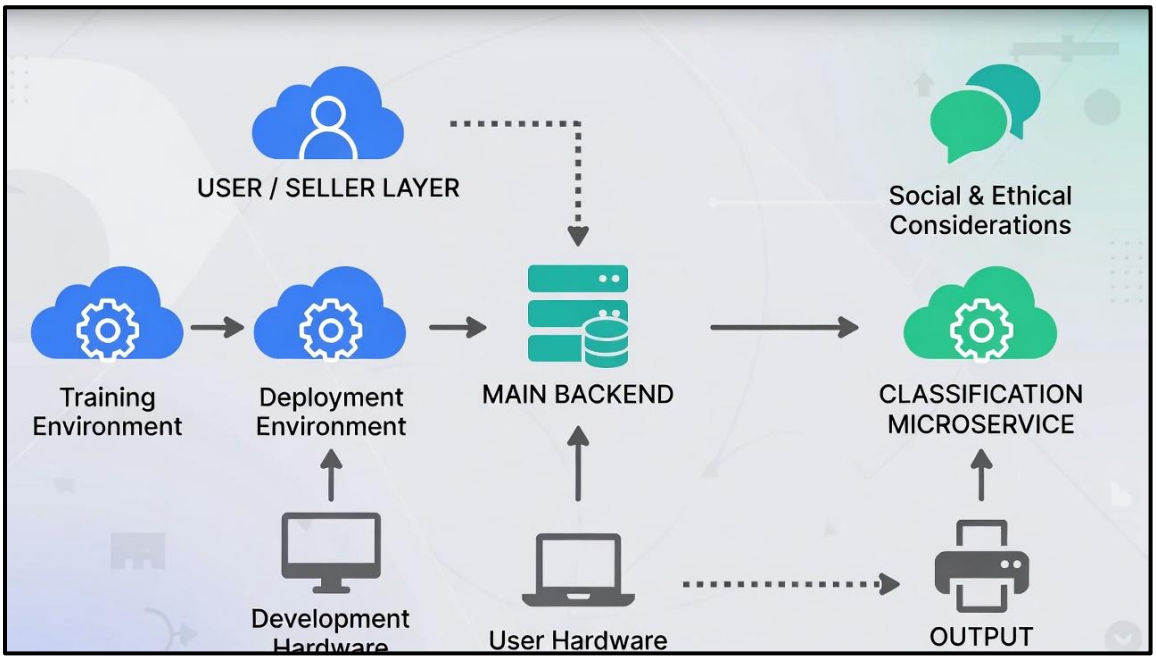


Figure 2: System Overview Diagram – Fabric Classification Module.

In order to facilitate picture categorisation inside a marketplace platform, the system employs a microservices architecture.

Parts:

- User Interface: The front end of the marketplace where users may post images.
- Classification API: Request-handling RESTful gateway.
- Images are preprocessed by the Image Processing Module (normalization, scaling).
- CNN Model: Uses inference to categorize pictures.
- All services are hosted by cloud infrastructure (cloud VMs/containers).
- Data Flow: User uploads a picture; Classification API; picture preprocessing; CNN model inference; JSON prediction delivered; Marketplace shows the outcome.
- Scalability: Dock erization and Kubernetes orchestration allow components to automatically scale in response to demand.
- Security: TLS encryption, rate limitation, input validation, and authentication are all enforced by the API gateway.

5. Requirements

5.1 Stakeholders

- Sellers (Manufacturers)
- Platform developers
- Waste management companies
- Textile recyclers

5.3 Functional Requirements

Table 2: Functional Requirements Table.

ID	Requirement Description
FR-01	The system shall accept image uploads in common formats (JPEG, PNG).
FR-02	The system shall pre-process uploaded images to a standardized size and format required by the CNN model.
FR-03	The system shall load and utilize the trained CNN model to perform inference on the pre-processed image.
FR-04	The system shall predict the fabric type from a predefined list of categories (Cotton, Polyester, Nylon, Viscose, Blend).
FR-05	The system shall predict the dominant color from a predefined list of categories (Red, Blue, Green, Yellow, Black, White, Other).
FR-06	The system shall predict the quality grade from a predefined list of categories (High, Medium, Low).
FR-07	The system shall return a confidence score (0.0 to 1.0) for each predicted attribute.
FR-08	The system shall return the predictions and confidence scores in a structured JSON format via a RESTful API endpoint.
FR-09	The system shall handle invalid inputs (e.g., unsupported file type, corrupt image) gracefully and return an informative error message.

FR-10	The system shall log all prediction requests, including timestamps and input image metadata (not the image itself, for privacy), for monitoring purposes.

1. Non-Functional Requirements

Table 3: Non-Functional Requirements Table.

ID	Requirement Description
NFR-01	**Accuracy:** The model's overall classification accuracy (all three attributes correct) on the held-out test set shall be >85%.
NFR-02	**Performance:** The API shall respond to a prediction request (including inference time) in under 2 seconds for at least 95% of requests.
NFR-03	**Scalability:** The microservice shall be designed to handle multiple concurrent requests (tested with up to 20 simultaneous users).
NFR-04	**Reliability:** The service shall have 99% uptime during development and testing phases.
NFR-05	**Usability:** The API shall be well-documented with clear examples, making integration straightforward for other developers.
NFR-06	**Maintainability:** The codebase shall be modular, well-commented, and version-controlled using Git. The model file should be easily replaceable with an updated version.
NFR-07	**Robustness:** The model should maintain reasonable accuracy (>70%) on images with slightly different lighting, angles, or backgrounds than the training set.

5.4 System Requirements

- Python 3.9+ is the programming language.
- Deep Learning Framework: PyTorch, Keras, or TensorFlow 2.x
- FastAPI is a web framework.
- Image processing: Pillow (PIL), OpenCV
- Deployment options include serverless container platforms (like Google Cloud Run), cloud virtual machines (like AWS EC2 t2. medium or a similar with a GPU option), and Docker containers.
- REST over HTTP/HTTPS is used for API communication.
- Storage: The microservice itself does not require a persistent database. Memory is used to process images. Listings and picture files are stored on the primary platform (e.g., in Cloudinary or MongoDB GridFS).
- Git (GitHub/GitLab) is used for version control.

5.5 User Requirements

- **Sellers (end users):**
    - To capture a clear picture of the fabric waste, they need to have access to a smartphone or a camera.
    - For optimal effects, the cloth should be placed on an uncluttered, contrasting backdrop (a straightforward suggestion in the user interface).
    - No technical knowledge of computer vision or artificial intelligence is necessary.
- **Platform Developers (Integrators):**
    - To send HTTP queries from the main backend, they must have access to the API documentation and base URL.

5.6 Use Cases

Table 4: Use Case 1 – Upload Waste Image for Classification Table.

Element	Description
Use Case ID	UC-01
Name	Upload Waste Image for Automated Classification
Summary	A seller uploads an image of a fabric waste batch, and the system automatically suggests a classification (type, color, quality) to streamline the listing creation process.
Priority	High
Preconditions	• The seller is logged into the marketplace platform. • The seller has navigated to the "Create New Listing" page.
Postconditions	• The system returns AI-generated classification suggestions, which are displayed to the seller. • The listing form is pre-filled with these suggestions.
Primary Actor(s)	Seller (Manufacturer)
Secondary Actor(s)	Marketplace Platform, Classification Microservice
Trigger	The seller clicks the "Upload Image" button and selects an image file.
Main Scenario	1. The seller selects an image file from their device. 2. The marketplace platform sends the image to the Classification Microservice's `/classify_fabric` endpoint. 3. The microservice validates, pre-processes, and runs inference on the image. 4. The microservice returns a JSON object with predicted attributes and confidence scores. 5. The marketplace platform receives the predictions and pre-fills the corresponding fields in the listing form (Fabric Type: Cotton, Color: Blue, Quality: High).

	6. The seller sees the pre-filled form and can proceed to add price and quantity.
Extensions	• **E1 (Low Confidence):** If the confidence score for any attribute is below a threshold (e.g., <0.7), the system highlights that field and asks the seller to review and confirm manually. • **E2 (Upload Error):** If the image is invalid or the API call fails, the system displays an error message ("Failed to analyze image. Please try again or enter details manually").
Open Issues	• What is the optimal confidence threshold for flagging predictions for human review? This may be determined during testing. • How to handle images with multiple fabric pieces of different types? Initial focus will be on single-type batches.

Table 5: Use Case 2 – Review and Correct AI Classification Table.

Element	**Description**
Use Case ID	UC-02
Name	Review and Correct AI Classification
Summary	After the AI provides its suggestions, the seller reviews them and has the option to manually correct any attribute before finalizing the listing.
Priority	Medium
Preconditions	UC-01 has been completed, and the listing form is pre-filled with AI suggestions.

Postconditions	The final listing is created with the seller-verified (and possibly corrected) attributes.
Primary Actor(s)	Seller (Manufacturer)
Secondary Actor(s)	Marketplace Platform
Trigger	The AI suggestions are displayed in the listing form.
Main Scenario	• The seller reviews the pre-filled fields (Fabric Type, Color, Quality). • If all suggestions are correct, the seller proceeds to fill in other details (price, quantity) and clicks "Publish." • If a suggestion is incorrect, the seller uses the dropdown menus to manually select the correct value. • The seller clicks "Publish," and the listing is saved with the corrected information.
Extensions	**E1:** The corrected data could be logged (anonymously) and used as additional training data to improve the model in future versions.
Open Issues	Should the system log user corrections? This could provide valuable data for future model retraining, but requires careful privacy considerations.

1. Wireframes

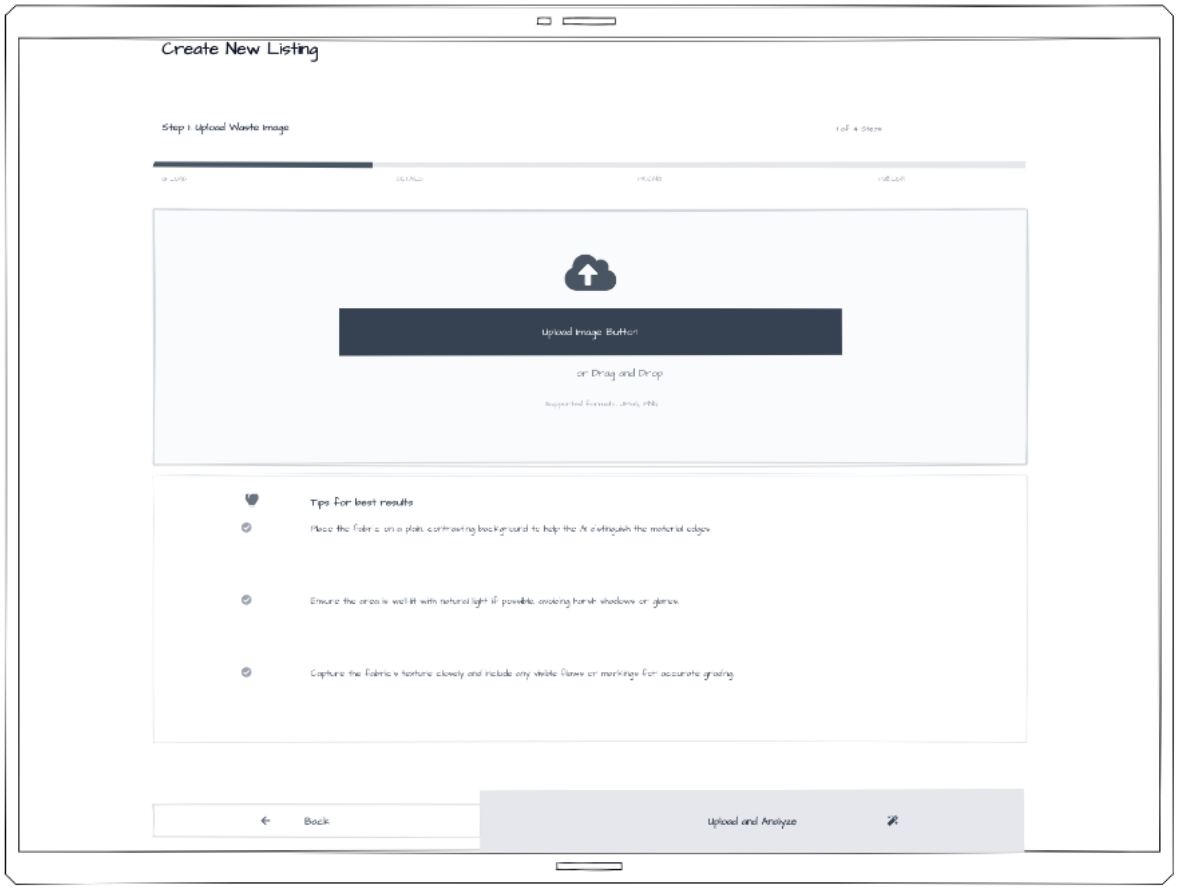


Figure 3: Wireframe – Image Upload Interface for Sellers.

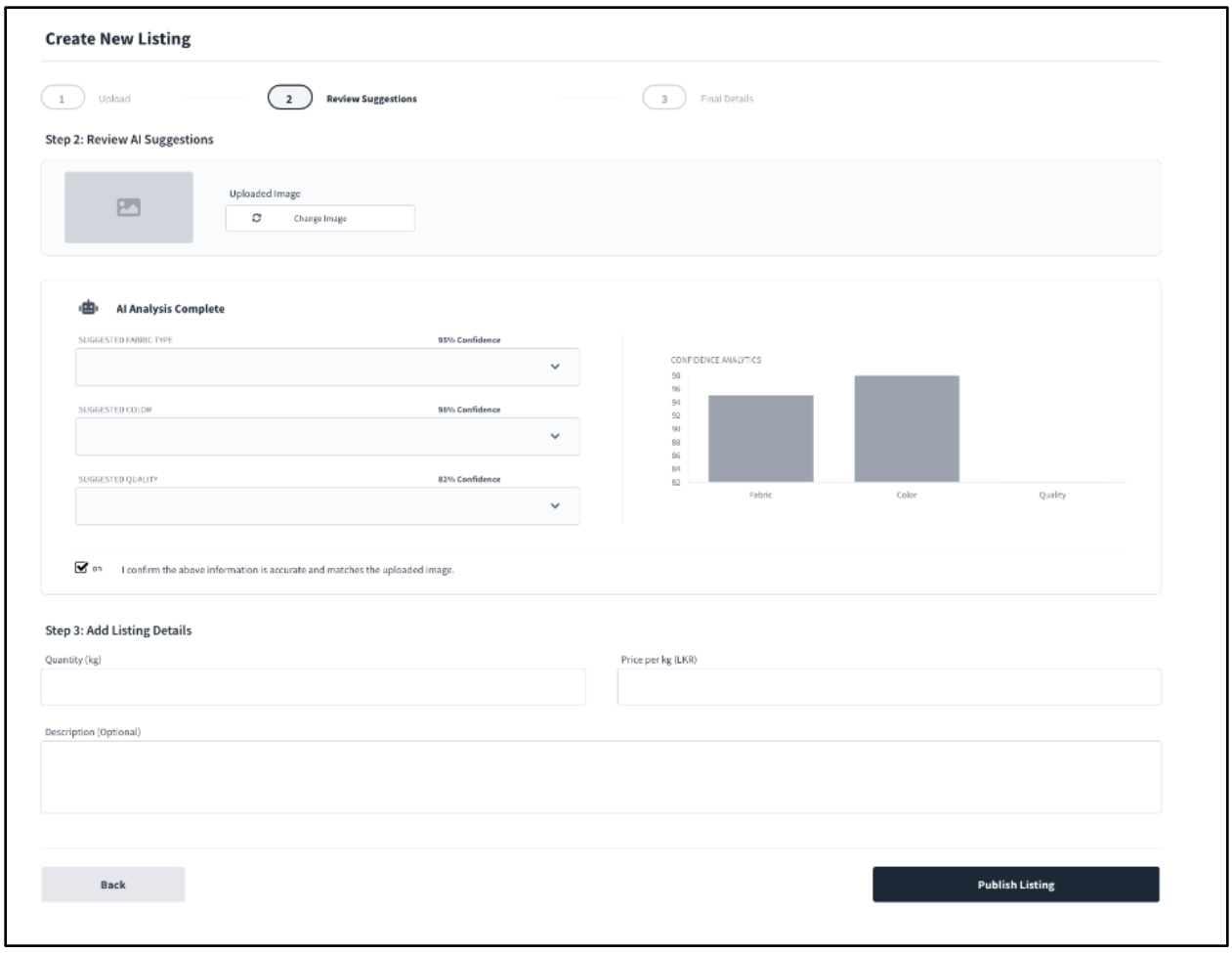


Figure 4: Wireframe – Classification Results Display.

6. Commercialization Plan

6.1 Target Market

- **Main**: This module is an essential component of the broader "TexTread" marketplace. Waste vendors (manufacturers) who list materials are their direct users.
- **Secondary**: A stand-alone "Waste Classification API" license for the module might be obtained for:
    - Waste management companies should automate their facilities' inventory and sorting processes.
    - Pre-sorting incoming textile waste streams is the purpose of recycling plants.
    - Big Manufacturers: For managing their own waste inventories.
- **Market Space:** The worldwide market for artificial intelligence (AI) in waste management and smart recycling is expanding quickly. The post-industrial textile waste in South Asia is the market segment that this module focuses on.

6.2 Value Proposition:

- For Sellers: Significantly cuts down on the time and effort needed to list garbage. increases consumer trust by offering standardized, AI-verified specs.
- For the Platform: Offers the fundamental data layer that makes all other intelligent features (effect analysis, pricing, and matching) possible. This is a crucial distinction from generic markets.
- **How to Promote:**
    - Demonstration Videos: Make short films that demonstrate how the system correctly classifies different sorts of garbage in a matter of seconds.
    - Collaborations with Manufacturers: To show the time-saving advantages and get input, provide free samples to nearby clothing companies.
    - Technical Publications: To build trust in the sustainable tech sector, publish a white paper on the model's accuracy and approach.

Model of Revenue

- API subscription (monthly tiered subscriptions or pay-per-call)
- SaaS approach for waste management firms
- Enterprise licensing for major factories and recycling facilities

Strategy for Pricing

- Limited free API calls per month under the freemium model
- Volume-based subscription levels (Basic, Professional, Enterprise)

An edge over competitors

- Classification of textile waste using specialized AI
- trained using data on post-industrial garbage in the region
- smoothly included into the TexTread marketplace environment

Intellectual Property

- CNN model architecture that is exclusive
- curated collection of photos of textile trash
- Codebase for the TexTread platform and categorization module

7. Budget and budget justification

Table 6: Budget Table.

Component	Amount (USD)	Amount (LKR)	Justification
Data Collection and Materials			
Fabric Waste Samples	10.00	3,000.00	Cost of obtaining different types of fabric waste samples from local tailors and small textile shops to build a representative dataset for model training.
Photography Setup (Lighting / Background)	5.00	1,500.00	Basic lighting setup and plain background cloth to ensure clear and consistent images when capturing fabric samples.
Cloud Computing Resources			
Cloud GPU Credits (Google Colab / Kaggle)	20.00	6,000.00	Used for training and testing deep learning models, such as CNN or transfer learning architectures.
Cloud Hosting for API (Basic VM Instance)	10.00	3,000.00	For hosting the prototype API during the system testing and demonstration stages.
Infrastructure and Tools			
Development Tools	0.00	0.00	All tools, such as Python, TensorFlow/PyTorch, VS Code, and GitHub, are open-source and freely available.
Internet Charges (Research & Development)	10.00	3,000.00	Internet access is required for research, downloading datasets, model training, and cloud platform usage.
Knowledge and Consultation			

Technical Consultation (Academic Guidance)	10.00	3,000.00	Occasional guidance from lecturers or domain experts regarding model architecture and training approaches.
Industry Consultation (Textile Knowledge)	5.00	1,500.00	Informal consultation with local textile professionals to validate fabric type identification and classification labels.
Contingency			
Miscellaneous / Contingency	5.00	1,500.00	Reserved for unexpected expenses such as additional samples or extended cloud usage.
TOTAL	**75.00**	**22,500.00**	The estimated total cost required to complete the project.

8. Work breakdown structure.

Table 7: Work Breakdown Table

Phase	Sub-Task ID	Task Description	Notes / Deliverables
1.1	**Research & Planning**		
	1.1.1	Conduct Literature Review (CNNs, Textile Classification)	Review paper summaries, references
	1.1.2	Define Classification Categories & Rubric	Final category list + labeling guideline
	1.1.3	Set Up Development Environment (Python, TF/PyTorch, Cloud)	Working Jupyter/Colab + cloud instance
1.2	**Dataset Engineering**		
	1.2.1	Source Fabric Waste Samples	Physical samples collected
	1.2.2	Capture Images (Controlled & Varied Conditions)	1,000–5,000+ raw images
	1.2.3	Label Images with Type, Color, Quality	Labeled dataset (CSV + folder structure)
	1.2.4	Implement Data Augmentation Pipeline	Augmentation code + sample visuals
	1.2.5	Create Train/Validation/Test Splits	Final splits (e.g. 70/15/15)
1.3	**Model Development**		
	1.3.1	Implement Baseline CNN (from scratch)	Simple ConvNet baseline
	1.3.2	Implement Transfer Learning Model (ResNet50/EfficientNet)	Pre-trained model loaded
	1.3.3	Train Initial Model (Frozen Base)	First training logs
	1.3.4	Fine-tune Model (Unfrozen Layers)	Improved performance
	1.3.5	Hyperparameter Tuning	Best hyperparameters + final candidate
1.4	**Model Evaluation**		

	1.4.1	Evaluate on Validation Set (Monitor Overfitting)	Learning curves, confusion matrix
	1.4.2	Final Evaluation on Test Set (Accuracy, Precision, Recall, F1)	Test set metrics table
	1.4.3	Error Analysis & Model Refinement	Qualitative error review + re-training
1.5	**API Development**		
	1.5.1	Set Up FastAPI Project Structure	Basic project scaffold
	1.5.2	Implement /classify_fabric Endpoint	Core inference route
	1.5.3	Integrate Trained Model for Inference	Model loading + prediction logic
	1.5.4	Implement Input Validation & Error Handling	Pydantic models, proper exceptions
	1.5.5	Write Unit & Integration Tests	≥70–80% coverage on critical paths
1.6	**Deployment & Integration**		
	1.6.1	Containerize with Docker	Dockerfile + docker-compose (optional)
	1.6.2	Deploy Microservice to Cloud VM	Running a container in the cloud
	1.6.3	Test API Integration with Main Platform	End-to-end working flow
	1.6.4	Document API Usage (Swagger/OpenAPI)	Auto-generated + supplementary guide
1.7	**Reporting & Documentation**		
	1.7.1	Write Methodology Section (Final Paper)	
	1.7.2	Write Results Section (Final Paper)	Tables, figures, comparison
	1.7.3	Prepare Project Demonstration	Live demo script + slides

9. Gantt chart

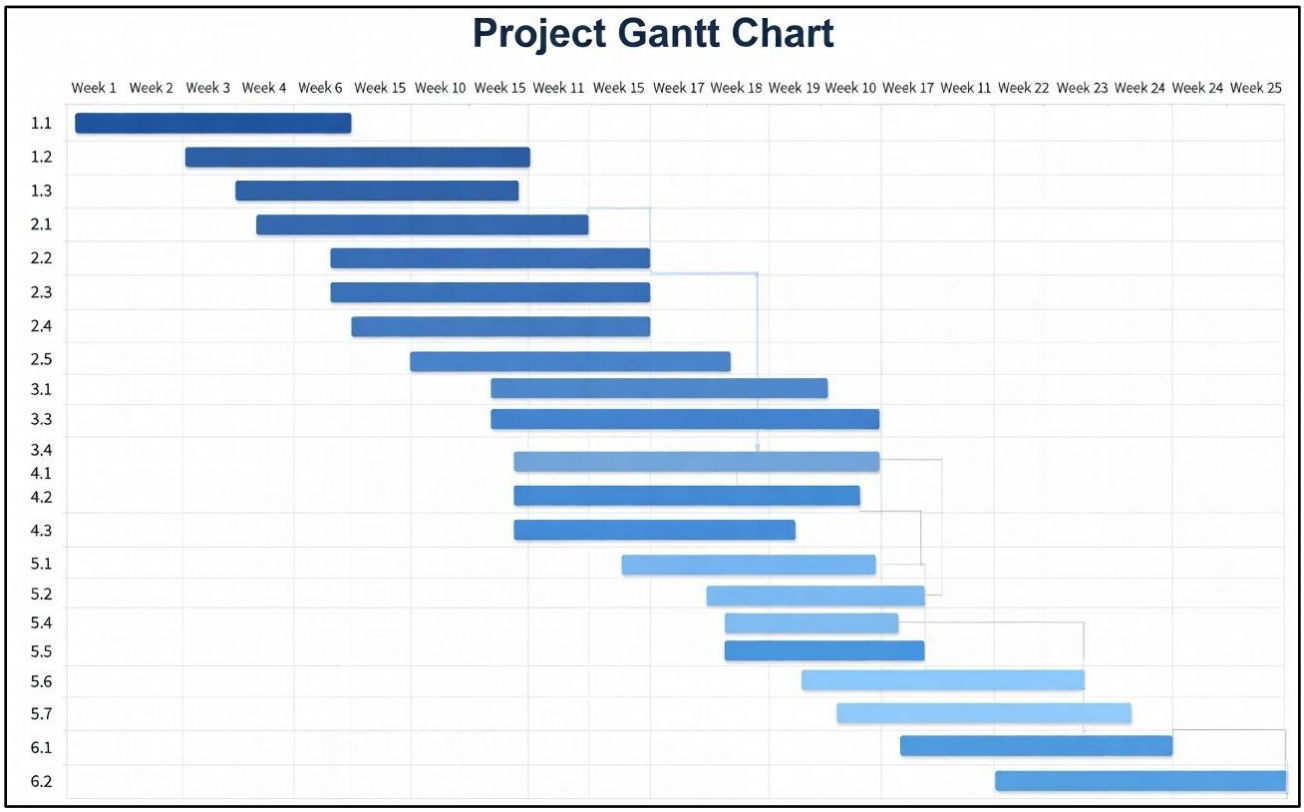


Figure 5: project Gantt Chart.

Table 8: Task Table

Task ID	Task Description	Start Week	End Week	Duration
1.1	In-depth literature review on CNNs for fabric classification	1	2	2
1.2	Finalize dataset categories and labeling rubric	2	3	2
1.3	Set up development environment and cloud resources	3	4	2
2.1	Source fabric waste samples from local suppliers/tailors	4	6	3
2.2	Photograph samples under varied conditions	5	7	3
2.3	Label images according to the rubric	6	9	4
2.4	Implement and apply the data augmentation pipeline	8	9	2
2.5	Split the dataset into train/validation/test sets	9	9	1
3.1	Implement baseline CNN model (simple architecture)	9	10	2
3.2	Implement transfer learning model (ResNet50/EfficientNet)	10	12	3
3.3	Train and tune model hyperparameters	11	14	4
3.4	Evaluate model performance on the validation set	13	14	2
4.1	Final evaluation on the held-out test set	14	15	2
4.2	Analyze errors and perform targeted refinements	15	16	2
4.3	Freeze and save the final model	16	16	1
5.1	Develop a FastAPI microservice with an inference endpoint	16	17	2
5.2	Implement input validation and error handling	17	18	2
5.3	Write unit tests for the API	18	19	2
5.4	Containerize the application with Docker	19	19	1
5.5	Conduct performance and load testing	19	20	2
5.6	Integrate with the main platform for end-to-end testing	20	21	2
5.7	Finalize API documentation and user guide	21	22	2
6.1	Prepare the final project report section	22	23	2
6.2	Prepare final presentation/demo	23	24	2

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Appendices

1. Example API Response

Below is an example JSON response returned by the classification API after processing a textile waste image.


```
{
  "fabric_type": {
    "label": "Cotton",
    "confidence": 0.92
  },
  "color": {
    "label": "Blue",
    "confidence": 0.87
  },
  "quality_grade": {
    "label": "Medium",
    "confidence": 0.81
  }
}
```